RITIK PATEL

Data Analyst & Machine Learning Engineer | Business Intelligence | Computer Vision | LLM Systems
Tempe, AZ (open to relocation) | ritikpatel@asu.edu | Available immediately
ritikpatel-asu.github.io/portfolio | linkedin.com/in/ritik-patel- | github.com/ritikpatel-asu

PROFESSIONAL EXPERIENCE

Data Analyst | WeAgile Software Solutions | Pune, India **Mar 2022 – Jun 2024**
Clients: The Home Depot, Wayfair, Ralph Lauren, Meta, Microsoft, Waste Management

- Owned daily KPI reporting for six enterprise retail and venue clients (footfall, dwell time, queue length, self-checkout, shrink), delivered to client operations teams in Power BI and Advanced Excel.
- Replaced four manual dashboards with one near real-time Power BI layer over a Python/Flask ETL pipeline, cutting report preparation time 60%+ and giving every client a single source of truth.
- Modeled the reporting layer across event, camera and site tables in SQL: reusable KPI views, window functions for rolling and period-over-period metrics, and validation queries and triggers that blocked bad writes, raising event quality 20%+.
- Evaluated production object-detection models (YOLOv7/v8, Faster R-CNN, SORT tracking) across 20+ camera feeds a day, tracked accuracy drift after camera-angle and store-layout changes, and triggered retraining when performance fell.
- Ran error analysis on missed and false detections, then retrained models and improved CVAT label quality, contributing to team results of mAP 0.71 to 0.84 and 38% fewer false positives at matched recall on The Home Depot self-checkout.
- Led client delivery as project lead: translated operations leads' requests into metric definitions and reporting requirements, ran Jira sprints and releases, and governed shared assets on AWS S3 and SharePoint.

Trainee Software Engineer | WeAgile Software Solutions | Pune, India **Nov 2021 – Mar 2022**

- Annotated ~500 images/day in CVAT with QA passes and inter-rater agreement checks, producing the training and evaluation sets for the team's detection models, and planned ~2 3D camera layouts a week (IP Video System Design Tool) for retail and venue deployments.

RESEARCH AND TEACHING EXPERIENCE

Research Assistant | ASU Speech/Language Genetics Lab | Tempe, AZ **May 2026 – Present**

- Building the analytics layer for an in-development reading app for users with dyslexia: progress and accuracy metrics per level and across the app, computed in Python and SQL on PostgreSQL.
- Prototyping Tableau dashboards that surface each user's level-by-level and overall performance, refining metric definitions as the app takes shape.

Graduate Teaching Assistant | ASU W. P. Carey School of Business | Tempe, AZ **Jan 2025 – May 2026**

- TA for three graduate courses, ~150 students each, 20 hrs/week. Assessed and coached students on SQL, statistical analysis, predictive modeling, deep learning, LLM fine-tuning, RAG, agentic workflows and prompt evaluation.
- Covered applied NLP topics including sentiment analysis, BERTopic topic modeling and multi-LLM integration across the curriculum.

PROJECTS

Financial RAG – Retrieval-Augmented QA over SEC Filings github.com/ritikpatel-asu/Financial-RAG-Pipeline-Retrieval-Augmented-QA

- Built hybrid-retrieval RAG over ~30 large-cap 10-K filings: BM25 + FAISS fused by Reciprocal Rank Fusion, BGE reranking, abstention gating and citation verification, served from FastAPI with Qwen2.5-7B and tracked in MLflow.
- Selected every component by benchmark (e5-base-v2 won a 4-model bake-off; reranking added +13% Recall@5) and reported the shortfalls: faithfulness 0.808, 7.5% hallucination, 4 of 7 injection attacks stopped, query rewriting shipped disabled.

Production ML Deployment – Steel Surface Defect Classification github.com/ritikpatel-asu/Production-ML-Deployment

- Shipped the full MLOps path for a 6-class defect classifier: MLflow registry with alias-based promotion behind validation gates, PSI and confidence drift signals, and retraining that leaves production untouched when a candidate fails; 34 tests, Terraform on AWS ECS/Fargate.
- Chose ResNet-18 over EfficientNet-B0 on measured CPU latency (17.1 vs 24.5 ms p95 at equal accuracy) and cleared a synchronous-inference bottleneck found in load testing: 134.7 to 193.4 req/s, p99 4,800 to 810 ms.

Retail Footfall & Operations Analytics Platform github.com/ritikpatel-asu/Retail-Footfall-Operations-Analytics-Platform

- Built an end-to-end analytics stack, object storage to DuckDB, PostgreSQL staging and dbt marts feeding Power BI, Streamlit and FastAPI, on 17 dbt models, 38 data tests and a ~2M-row incremental fact table.
- Defined a governed KPI dictionary (footfall, 7-day rolling traffic with WoW/MoM change, camera availability) with a z-score anomaly baseline; the latest partition scored 92.1% data quality against a 90% gate.

Multimodal Wildfire Detection – M.S. Capstone github.com/ritikpatel-asu/wildfire-multimodal

- Fused a ResNet vision model, a fine-tuned DistilBERT text model and geospatial signals (spaCy NER, haversine distance, temporal decay) over 75,709 NASA FIRMS detections in 324 events; ablation showed fusion lifted ROC-AUC from 0.556 to 0.789.
- Found and fixed two data-leakage bugs before trusting any result, near-duplicate tiles across splits and template leakage that inflated text accuracy to 100%, restoring an honest 85.3%; 116 tests, MLflow, FastAPI.

Network Intrusion Detection with ML – Co-authored Research Paper github.com/ritikpatel-asu/NIDS-ML_Paper

- Benchmarked seven classifiers, including Random Forest, XGBoost, SVM and KNN, on NSL-KDD and CIC-IDS2017 (~690k records) for multi-class attack typing; Random Forest led at 99.77% binary and 99.71% multi-class, while macro-F1 of 0.69–0.94 exposed weak recall on rare attack classes.
- Cut 41 features to the 30 most predictive with ExtraTrees and used one fixed 80/20 split so binary and multi-class runs stayed comparable.

SKILLS

Data & Analytics: SQL (Window Functions, CTEs), Python (Pandas, NumPy), PostgreSQL, DuckDB, Power BI (DAX), Tableau, Advanced Excel
Analytics Engineering: dbt, ETL Pipelines, Data Modeling, Data Quality Testing, KPI Definition, Statistical Analysis, Anomaly Detection
ML & AI: PyTorch, Scikit-learn, XGBoost, Hugging Face, YOLO, Faster R-CNN, DistilBERT, spaCy, RAG (BM25, FAISS, Reranking)
MLOps & Tools: FastAPI, Flask, Streamlit, MLflow, Docker, Terraform, AWS (S3, ECS/Fargate), pytest, Git, Jira, CVAT

EDUCATION AND CERTIFICATIONS

M.S. Data Science, Analytics and Engineering | Arizona State University, Tempe, AZ | GPA 3.61 **May 2026**
Coursework: Statistical ML, Deep Learning, Computer Vision, NLP, Large Language Models, Data Mining, Database Management, Data Visualization

Master of Computer Applications | Savitribai Phule Pune University, India | GPA 9.7/10 **May 2022**

Certifications: Google Data Analytics | IBM Data Science | DeepLearning.AI Deep Learning Specialization | IBM RAG and Agentic AI